EE1007
Introduction to Computer Science Laboratory

Chapter 0
Course Introduction

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Course Website

◆國立中央大學新 ee-class 數位學習平台
➢https://ncueeclass.ncu.edu.tw/



Be Sure to Update Your Email…



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Outline

- Basic information about this course
- What will you learn in this semester?
- Grading and regulations



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Introduction to Computer Science Laboratory (EE1007A/B)

- This is Lab course for “Introduction to Computer I (EE1003)”
- Course type
  - Undergraduate course
  - Department Required Course
  - Freshman students
- Credits: 1 unit
- Class Time: Thursday 18:00-20:50
- Class Location
  - Group A: I-002 (9/23: meet.google.com/yyc-avxi-ydi)
  - Group B: I-202 (9/23: meet.google.com/gkq-qoho-xgx)

About Virtual Classes…

- Please use your real name in your Google Account
  - Otherwise TA may be not able to find you



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  - Otherwise TA may be not able to find you



About Virtual Classes…

- If you have concerns on putting your name on Google Account……
  - Please apply an new account of NCU G Suite for Education
  - https://www.cc.ncu.edu.tw/page/Gsuite



Instructor

- Name：陳聿廣 (Andy Chen)
- Affiliation：Assistant Professor, Department of EE, NCU
- Experience：
  - Adjunct Assistant Professor, Department of CS, NTHU
  - Assistant Professor, Department of CSE, YZU
  - Lecturer, Department of ECE, Missouri S&T, Missouri, USA
  - Research Fellow, Department of CSE, Notre Dame, Indiana, USA
- Research Interests：
  - Electronic Design Automation (EDA)
  - Circuit Aging Mitigation and Tolerance
  - Reliable IMC Design
  - PDN Design and Optimization
  - AI Accelerator Design for Edge Devices
  - Hardware Security

Instructor

- Email: andyygchen@ee.ncu.edu.tw
- Phone: 03-4227151#34457
- Office: E1-336, Engineering Building 2
  (College of Electrical Engineering & Computer Science)
- Office hours:
  - Wednesday 3PM-5PM
  - I'm often available at other times
    - Please reserved by E-mail



Teaching Assistants

- TA Office: E1-359
- Office Hours: Reserved by E-mail (A:金昌明 B:楊云緯)

A 班助教

黃柏燁　梁育銓　林俐秀　金昌明

B 班助教

王啟旭　何宜真　楊云緯　楊修宜

Course Purpose

- A good opportunity for practicing and improving your C++ coding skills
- Why we need C++ coding skills?
  - To solve problems through programming
  - Fundamental skills for learning programming
  - Can help you deal with complicated problems in other courses or competitions
  - Because it is a required course……



Coding Environment (IDE)

- IDE: Code::Blocks 20.03
- Compiler: GCC/G++/GFortran compiler and GDB debugger from MinGW-W64 project (version 8.1.0, 32/64 bit, SEH).
- C++ version: C++ 11 as basis



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About this course…



- A good opportunity for students to practice C++ programming
- Focus on writing a "good" C++ code
  - Not for competition, but for large project in the future
- Majorly focus on basic programming skills
  - Not the fancy things
- Students are expected to take Introduction to Computer I (EE1003) course at the same time





A good C++ code?



- Working Code
  - Looks good to me
  - Solve today's problem
  - Easiest to measure from outside
- Good Code
  - Looks good to you
  - Solve tomorrow's problem
  - Evaluating it is just as hard as writing it
  - Full of comments



Course Objectives

- You are in the wrong class if you
  - want to practice more on fancy skills of C++
  - want to participate time-limited programming competitions (ex:ICPC)
  - are not interested in the programming at all
    - But this is a required course Orz…





Contents

- Chapter 1 - Introduction to Computers
- Chapter 2 - Introduction to C++ Programming
- Chapter 3 - Control Statement: Part 1
- Chapter 4 - Control Statement: Part 2
- Chapter 5 – Functions and Recursion
- Chapter 6 - Arrays and Vectors
- Chapter 7 - Pointers
- Chapter 8 - Sequential-Access Files
- Chapter 22 - Bits, Characters, C Strings and structs
- Chapter 9 - Classes
- Chapter 10 - Classes: A Deeper Look
- Chapter 15 - Stream Input/Output

Syllabus

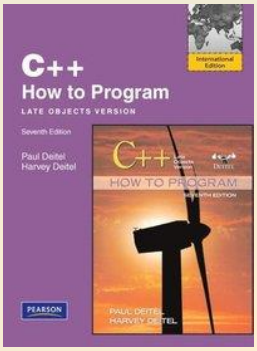
Date	Week	Progress	Note
9/16	1	Course Introduction (Pre-recorded)	
9/23	2	Lab 1: Your first program	Virtual Class
9/30	3	Lab 2: Variables and arithmetic	
107	4	Lab 3: Control statement I (flow control)	
10/14	5	Lab 4: Control statement II (repeat control)	
10/21	6	Lab 5: Control statement II (repeat control)	
10/28	7	Lab 6: Functions	
11/4	8	Lab 7: Recursion	
11/11	9	Midterm Exam (on-line programming exam)	

Syllabus

Date	Week	Note	Note
11/18	10	Lab 8: Arrays	
11/25	11	Lab 9: Vector and string	
12/2	12	Lab 10: Pointer I	
12/9	13	Lab 11: Pointer II	
12/16	14	Lab 12: Files and parser	
12/23	15	Lab 13: Bits, Characters, C Strings and structs	
12/30	16	Lab 14 : Class I	
1/6	17	Lab 15: Class II	
1/13	18	Final Exam (on-line programming exam)	

Textbook

- C++ How to Program - Late Objects Version
  - By P. J. Deitel and H. M. Deitel
  - 7th Edition (2011)
  - Pearson Education Inc.
  - ISBN: 0-13-248458-7



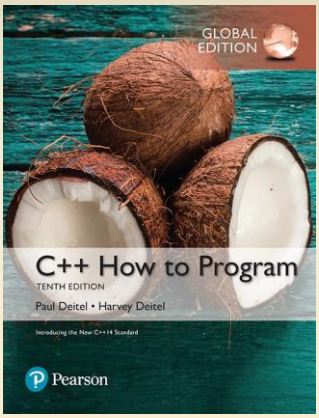
Textbook

- Handouts from Instructor
- The slides will be posted on ee-class prior to the class
- For class attendees only
  - Please don't distribute without prior permission



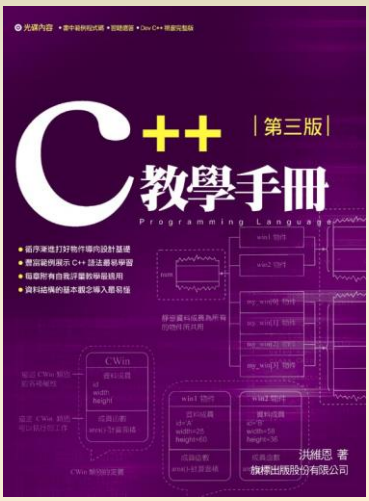
Reference Books

- C++ How to Program
  - By P. J. Deitel and H. M. Deitel
  - 10th Edition
  - Pearson Education Inc.
  - ISBN-10: 013444888X



Reference Books

- C++ 教學手冊 第三版
  - 洪維恩
  - 旗標出版
  - ISBN：9789577179371



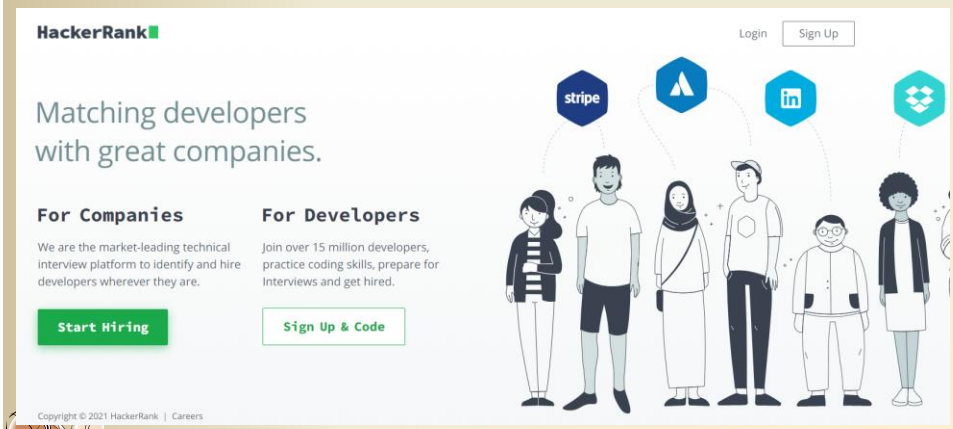
Resource

◆ZeroJudge



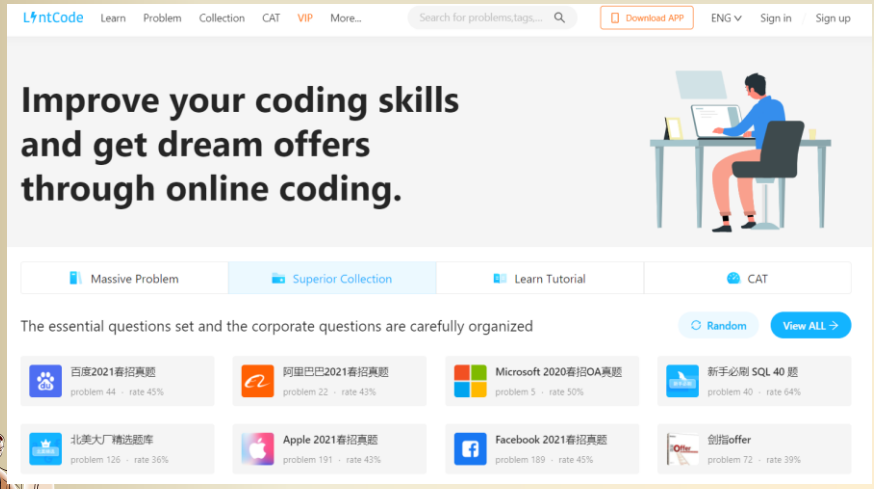
Resource

◆HackerRank



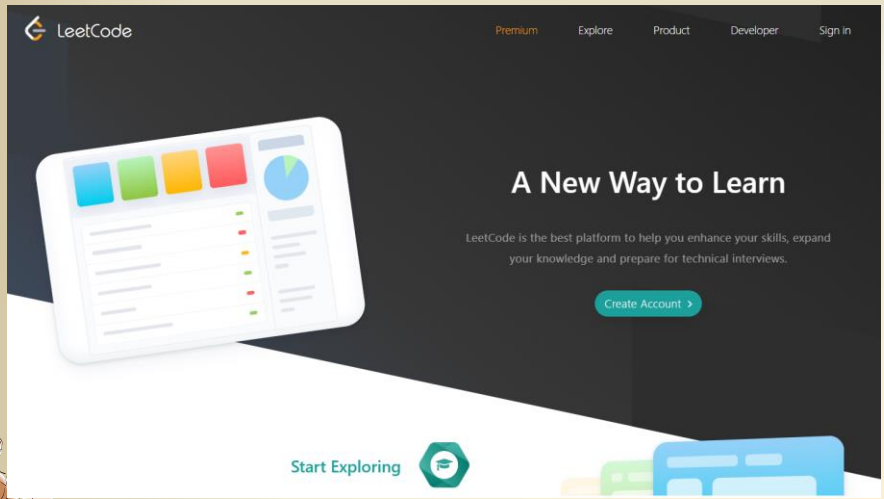
Resource

◆LintCode



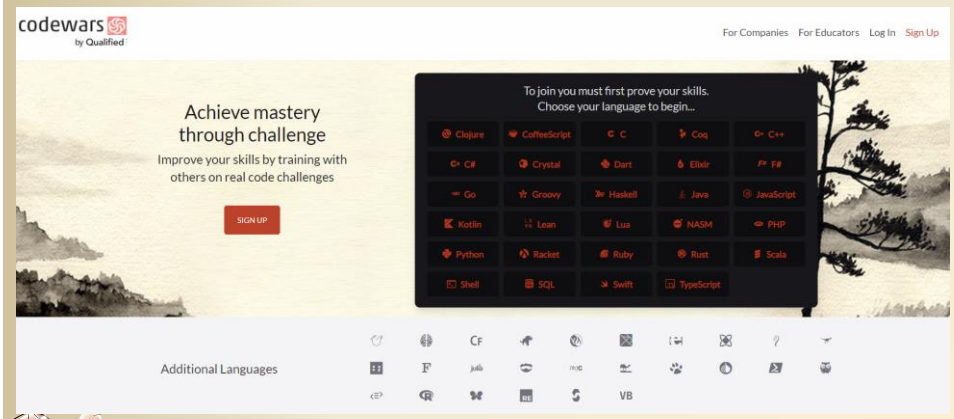
Resource

◆LeetCode



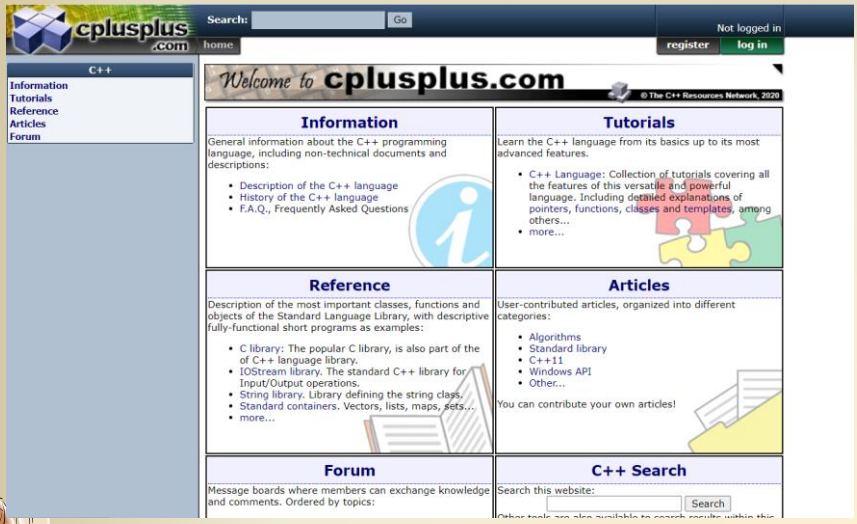
Resource

◆CodeWars



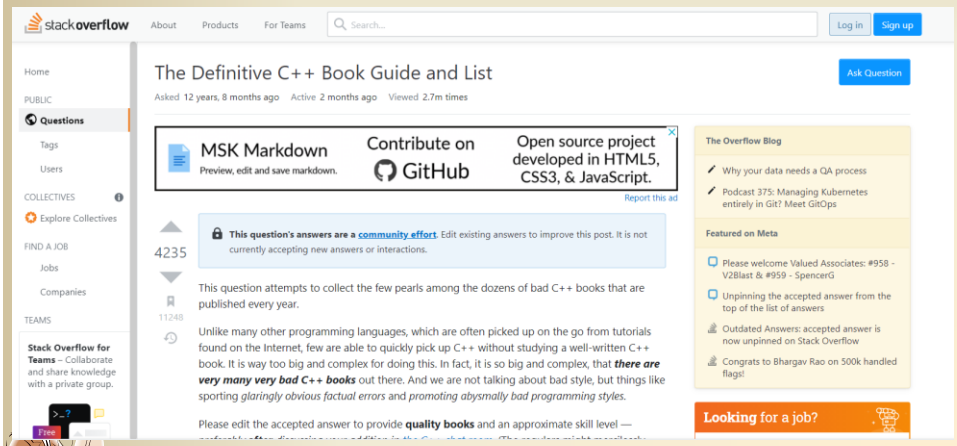
Resources

◆cplusplus.com



Resources

- Stack overflow
  - https://stackoverflow.com/questions/388242/the-definitive-c-book-guide-and-list



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Grading Policy

Lab Assignment	60%
Midterm Exam	10%
Final Exam	20%
Attendance	10%
Bonus Points	??%

- You will pass this course if you get 60 points or more
- I only can give you at most 100 points

Lab Assignment (60%)

- In each lab, we will provide 1~3problems (relatively easy coding problems) over the semester
- Students have to finish the coding individually in the class
- You have to find a TA to demo (check your code) and upload your code to ee-class before you leave the classroom



Lab Assignment (60%)

- Please finish the lab assignment in the class (BEFORE 21:00 each Thursday)
- Grading of each lab will be announced after each class through ee-class
- Late submission penalty
  - Make-up within 72hrs: 80%
  - More than 3 days: 0 point
  - You have to find a TA for making-up demo at the following week, otherwise you will get 0 point



Midterm(10%)/Final Exams (20%)

- On-line programming exam (individual)
- 3~5 coding problems in 3 hours
- You can bring anything but no internet-connectable devices
- Details will be announced later

Attendance (10%)

- You are required to attend the class (fix seat)
- Ask for absence with reasonable excuse through E-mail (to A班:梁育銓助教/B班:王啟旭助教 and cc to me) BEFORE class, otherwise you get 0 for the Lab of the week.
  - The classes will start at 18:00, you should ask for absence before that time!
- If you cut the class, you will be responsible for procuring any material, information, handouts, or announcements that you missed

Bonus (??%)

- We may have some bonus labs which provide you extra points
- You are not penalized for missing the bonus labs
- To obtain higher score, I strongly recommend you to take them when available





How to Pass This Course



- Attend every class
  - Concentration
  - **Ask**, if you do not understand (in class or office hours)
  - **Finish every problem in the class**
- Study
  - Preview, review, and practice
  - Focus on handouts and use the textbook as a reference
- You are expected to spend a lot of time on debugging
- Do good job at exams
- Make good friendships with your classmates / TAs (?)

How to Pass This Course

- Cheating?!?
  - **Don't do it!**
  - My policy is to fail you in the course and report your behavior to the University.

Academic Honesty

- "Cheating" is very uncivilized behavior and should be avoided at all cost
- **Oral discussion is encouraged and is not considered as cheating**
- Copying someone's codes / exams or part of codes / exams is cheating
- If cheating is discovered
  - My policy is to fail all of you (抄襲者與被抄襲者) in the course and report your behaviour to the University
  - All students involved will be reported

Academic Honesty

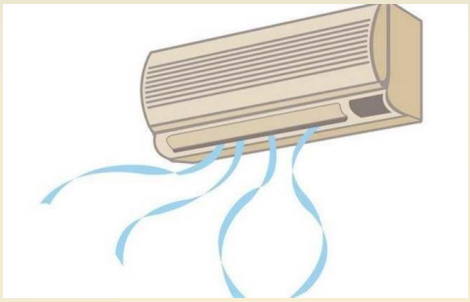


How to Pass This Course

- Cheating including……
  - Copy other's code
  - Ask others to write a code for you
  - Copy and paste from internet resources without clearly identify
  - ……

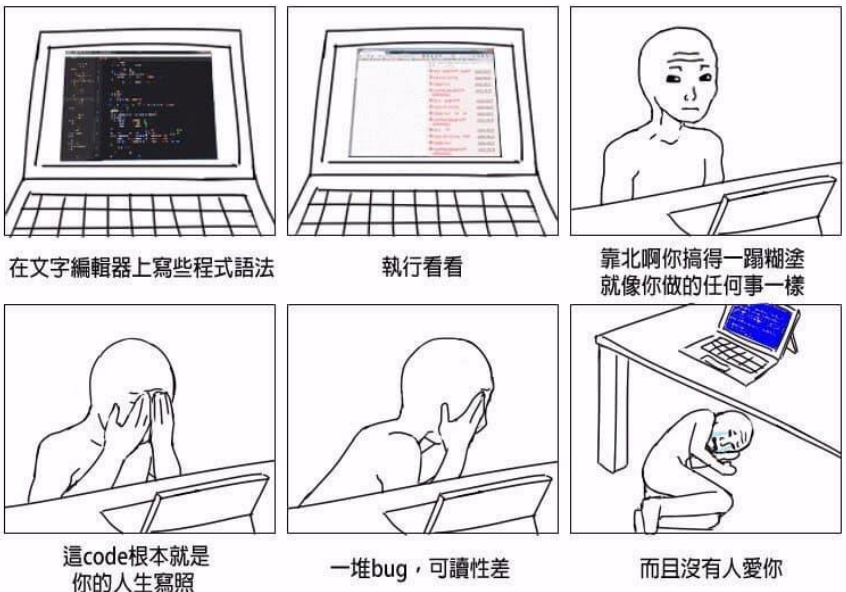
甜度?涼度?

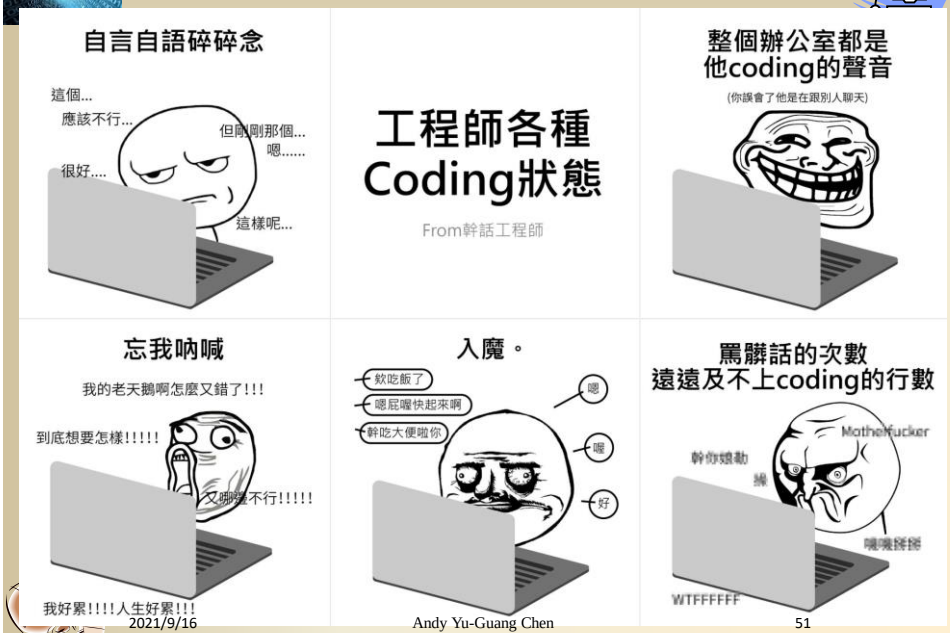




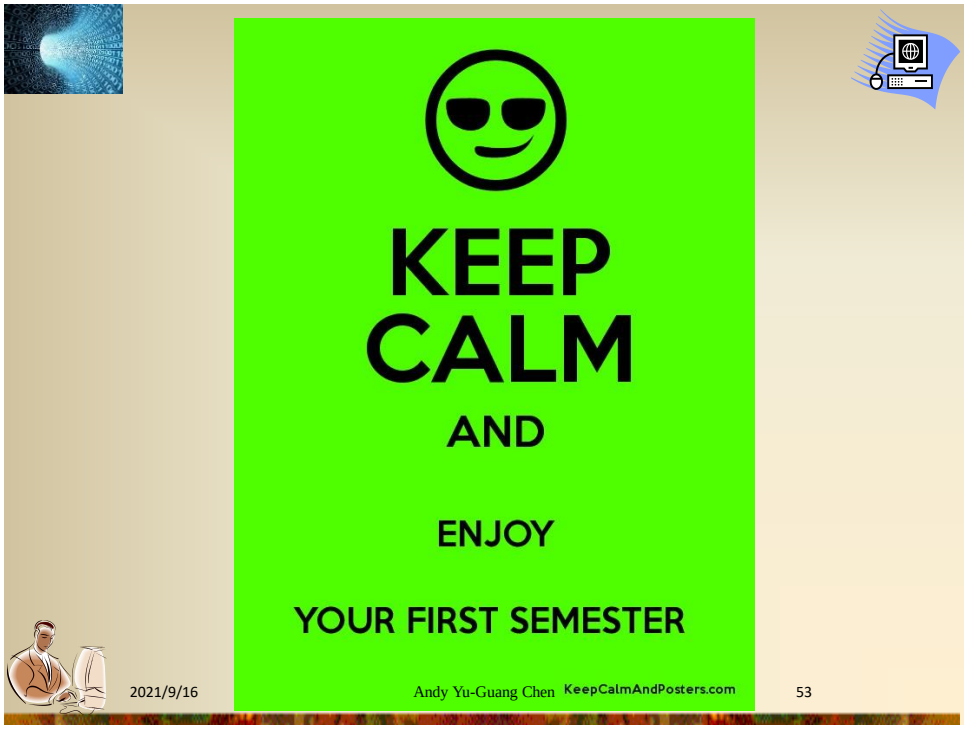

READY
FOR
DEPARTURE

如何coding


在文字編輯器上寫些程式語法
執行看看
靠北啊你搞得一蹋糊塗
就像你做的任何事一樣
這code根本就是
你的人生寫照
一堆bug，可讀性差
而且沒有人愛你


自言自語碎碎念
這個...
應該不行...
但剛剛那個...
嗯.....
很好....
這樣呢...
工程師各種
Coding狀態
From幹話工程師
整個辦公室都是
他coding的聲音
(你誤會了他是在跟別人聊天)
忘我吶喊
我的老天鵝啊怎麼又錯了!!!
到底想要怎樣!!!!!
又哪裡不行!!!!!
我好累!!!!人生好累!!!
入魔。
欸吃飯了
嗯屁喔快起來啊
幹吃大便啦你
嗯
嗯
好
罵髒話的次數
遠遠及不上coding的行數
MotherFucker
幹你娘勒
操
嘎嘎揉揉
WTFFFFFF


It doesn't work..... why?
It works..... why?
MemeBlender.com



Q&A



- Please leave you question(s) at the discussion thread “Course Introduction” at the ee-class





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